

Chapter 2: SAMPLING OF CONTINUOUS-TIME SIGNALS

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Problem 1

Given a continuous signal

$$x(t) = 2 \cos(8\pi t) + 2 \cos(4\pi t) + \cos(6\pi t), \quad [t] = \text{ms}, \quad f_s = 5 \text{ kHz}.$$

Find the aliasing signal $x_a(t)$.

Problem 2

A continuous signal

$$x(t) = 4 \cos(2\pi t) \cos(8\pi t) \cos(12\pi t), \quad [t] = \text{s}, \quad f_s = 10 \text{ Hz}.$$

Find the aliasing signal $x_a(t)$ and show that the sampled signals of $x(t)$ and $x_a(t)$ are the same (prove that $x[n] = x_a[n]$).

Problem 3

A DSP application requires $X_m = 10 \text{ V}$, quantization errors $e_{\text{rms}} < 1 \text{ mV}$.

1. Find the number of bits B used by the ADC converter.
2. Determine the actual e_{rms} .
3. Find the SNR (in dB) of the converter.

Hint:

$$e_{\text{rms}} = \frac{\Delta}{\sqrt{12}}, \quad \text{and} \quad \text{SNR} = 6.02B \text{ (dB)}$$

Problem 4

A music song in CD (2 channels) has the following parameters:

- Sampling frequency: $f_s = 44.1$ kHz
- Quantization: $B = 16$ bits

Assume the song duration is 5 minutes.

1. Determine the average capacity (in MB) of that song.
2. How many songs can be stored on a 600 MB CD?

Problem 5

A signal has bandwidth of 15 kHz.

- (a) Find the Nyquist rate.
- (b) If quantization has $L = 65,536$ levels, how many bits can encode 1 sample?
- (c) What is the bit rate for this signal?